

Notes: Pedersen (JFE, 2022)

Game on: Social Networks and Markets

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1 Big picture

- **Question.** How do social networks, influencers, fanatic investors, and rational traders affect asset prices, beliefs, portfolios, trading volume, momentum, reversals, bubbles, and crashes?
- **Setting.** A discrete-time asset market with one risky asset. Investors communicate through a social network and trade before the uncertain part of the fundamental value is revealed.
- **Core friction.** Some investors learn through repeated social averaging. Because the same idea can echo through the network many times, naive investors can overreact to repeated information.
- **Four types of investors.** The model contains naive investors, fanatics, rational long-term investors, and rational short-term investors.
- **Fanatics.** Fanatics are stubborn about their own views. They listen only to themselves and can represent extremely committed investors, meme traders, or people spreading a false view because they believe it.
- **Rational investors.** Rational investors initially listen to everyone and recover the rational view. After that, they become stubborn about the truth because repeated views contain no new information.
- **Main conceptual contribution.** The paper embeds a DeGroot-style social learning network into an asset-pricing model and solves in closed form for beliefs, portfolios, and prices.
- **Influencers versus thought leaders.** Influencers are people whose attention choices move many others. Thought leaders are hardheaded agents whose views survive in the long-run average opinion.
- **Main mechanism.** An idea starts with fanatics, spreads through naive investors, is amplified by influencers, affects trading demand, and appears in prices.
- **Main price result.** Price equals a rational benchmark plus a social-network component. In the long run, the network component equals fanatic mispricing weighted by fanatic thought leadership.
- **Main trading result.** Rational short-term traders may initially ride a bubble because they understand that naive demand will keep building. Later, they bet on reversal when the price is too far from fundamentals.
- **Main implications.** The model generates bubbles, social spillovers, excess volatility, spikes in trading volume, price momentum, fundamental momentum, long-run reversal, value effects, and local-bias-style portfolio patterns.
- **Application.** The GameStop episode is used as an illustrative case because it displays social-media attention, extreme views, influencer amplification, high volume, high volatility, and sharp price movements.
- **Takeaway.** In a social market, influence is not only about wealth. Market impact can also come from who listens to whom.

2 Data and measurement (key objects)

2.1 Why this is mainly a theory paper

- This is not a standard empirical paper with a dataset, regression table, natural experiment, or instrument.
- The central object is a formal model of social learning and asset-market equilibrium.
- The GameStop discussion is a case study that illustrates the model rather than a formal causal test.

Interpretation. The paper should be read as a benchmark theory of how narratives, social communication, and network position can matter for asset prices.

2.2 Asset value, signals, and revelation

The asset has supply s and fundamental value:

$$v + u(t), \tag{1}$$

where $u(t)$ is publicly observed and follows a random walk, while v is initially unobserved.

Each investor i starts with a signal:

$$x_i(0) = v_i. \tag{2}$$

Collectively, all signals reveal the unknown component:

$$v = \sum_{i=1}^N \kappa_i v_i, \quad \sum_{i=1}^N \kappa_i = 1. \tag{3}$$

The value v is revealed at random time τ . Conditional on no previous revelation, revelation occurs each period with probability π .

Before revelation, price is $p(t)$. After revelation, price equals fundamental value:

$$\text{price}(t) = p(t)\mathbf{1}_{t < \tau} + [v + u(t)]\mathbf{1}_{t \geq \tau}. \tag{4}$$

Interpretation. The model separates public observable value $u(t)$ from an unknown component v . A bubble can persist only while v has not been revealed.

2.3 Investor groups

The number of investors is:

$$N = N_n + N_f + N_l + N_s, \tag{5}$$

where N_n is the number of naive investors, N_f fanatics, N_l rational long-term investors, and N_s rational short-term traders.

- **Naive investors** repeatedly update by listening to their network.
- **Fanatics** listen only to themselves and keep their initial view.
- **Rational long-term investors** learn the truth and trade based on value at revelation.
- **Rational short-term investors** learn the truth but trade based on expected next-period returns.

2.4 The social network

The network is summarized by an adjacency matrix A . Row A_i gives the weights investor i places on the views of others. A naive investor updates as:

$$x_i(t+1) = A_i x(t). \quad (6)$$

The matrix can be partitioned as:

$$A = \begin{pmatrix} A_{nn} & A_{nh} \\ 0 & I \end{pmatrix}, \quad (7)$$

where h denotes hardheaded agents: fanatics plus rational investors after time 1.

Interpretation. Naive investors keep listening to others. Hardheaded investors keep their own views. This is the central structure that creates persistent disagreement.

2.5 Wealth and risk-bearing capacity

Investor i has wealth or absolute risk tolerance w_i . Total wealth is:

$$w. = \sum_{i=1}^N w_i. \quad (8)$$

Interpretation. Wealth gives direct price impact through demand. Network position gives indirect price impact through belief transmission.

2.6 Key endogenous outcomes

The main objects solved in the paper are:

- investor beliefs $x_i(t)$,
- wealth-weighted average beliefs $\bar{x}(t)$,
- influencer values μ_i ,
- thought leadership weights θ_j ,
- investor demands $d_i(t)$,

- price $p(t)$,
- the rational price $p_r(t)$,
- the network price component $p_n(t)$,
- volume, volatility, value, momentum, and reversal.

3 Models and empirical specifications (all equations + interpretation)

3.1 Rational view

A rational investor listens to everyone initially and computes:

$$x_r = \mathbb{E}[v \mid x_1(0), \dots, x_N(0)] = (\kappa_1, \dots, \kappa_N)x(0) = v. \quad (9)$$

Interpretation. Rational investors recover the truth from the full set of initial signals. They then ignore repeated echoes because repeated information is not new information.

3.2 Belief dynamics with hardheaded agents

Using the block matrix for A , beliefs satisfy:

$$x(t) = A^{t-1}x(1) \rightarrow \begin{pmatrix} (I - A_{nn})^{-1}A_{nh}x_h \\ x_h \end{pmatrix}. \quad (10)$$

Interpretation. Naive views converge to weighted averages of hardheaded views. If the only hardheaded agents are rational, truth dominates. If fanatics are also hardheaded, their views survive.

3.3 Long-term investor demand

Naive, fanatic, and rational long-term investors choose demand:

$$d_i(t) = \frac{\pi w_i}{\sigma_u^2} [x_i(t) + u(t) - p(t)]. \quad (11)$$

Interpretation. Demand is larger when perceived value exceeds price, when the investor has more risk-bearing capacity, and when revelation is expected sooner.

3.4 Short-term investor demand

A rational short-term trader chooses:

$$d_i(t) = \frac{w_i}{\sigma_u^2} [(1 - \pi)\mathbb{E}_t p(t+1) + \pi(x_r + u(t)) - p(t)]. \quad (12)$$

Interpretation. Short-term traders care about the next-period expected price. They may buy today even when the asset is overvalued if they expect social-network demand to keep pushing prices up.

3.5 Market clearing

The market-clearing condition is:

$$s = \sum_{i=1}^N d_i(t). \quad (13)$$

Interpretation. Social beliefs matter for prices because they affect demand, and price is the object that clears the asset market.

3.6 Influencer values

The wealth-weighted average opinion is:

$$\bar{x}(t) = \sum_{i=1}^N \frac{w_i}{w} x_i(t) = \frac{1}{w} w' x(t). \quad (14)$$

Influencer values of naive agents are:

$$\mu' = \frac{w'_n}{w} (I - A_{nn})^{-1}. \quad (15)$$

Interpretation. The inverse $(I - A_{nn})^{-1}$ is an echo matrix. It counts how a view moves from one naive investor to others directly and indirectly through the network.

3.7 Thought leadership

Thought leadership of hardheaded agents is:

$$\theta'_h = \mu' A_{nh} + \frac{w'_h}{w}. \quad (16)$$

If naive agent i increases his following of hardheaded agent j by ε , then:

$$\frac{\partial \theta_j}{\partial \varepsilon} = \mu_i. \quad (17)$$

Interpretation. A hardheaded agent becomes a thought leader when influential naive investors listen to him. An influencer matters because his attention choices shift the long-run importance of hardheaded views.

3.8 Rational price benchmark

If wealth is held only by rational investors, the price is:

$$p_r(t) = x_r + u(t) - \frac{s\sigma_u^2}{\pi w}. \quad (18)$$

Interpretation. The rational price is expected fundamental value minus a risk premium. This is the clean benchmark without social-network distortions.

3.9 Network price component

In the full model:

$$p(t) = p_r(t) + p_n(t), \quad (19)$$

where:

$$p_n(t) = \frac{w_n(1-c)}{w} \sum_{k=0}^{\infty} c^k [\bar{x}_n(t+k) - x_r] + \frac{w_f}{w} (\bar{x}_f - x_r), \quad (20)$$

and:

$$c = \frac{1}{1 + \frac{w_s}{w} \frac{\pi}{1-\pi}}. \quad (21)$$

As time goes to infinity:

$$p_n^\infty = \sum_{j=N_n+1}^{N_n+N_f} \theta_j (x_j - x_r). \quad (22)$$

Interpretation. Long-run social mispricing is fanatic optimism or pessimism multiplied by thought leadership. A wrong view matters most when the network gives it a large audience.

3.10 Fanatic and influencer effects on price

The long-run price sensitivity to fanatic j 's valuation is:

$$\frac{\partial p_n^\infty}{\partial x_j} = \theta_j. \quad (23)$$

If naive investor i with influencer value μ_i increases following of fanatic j at the expense of a rational agent, then:

$$\frac{\partial p_n^\infty}{\partial \varepsilon} = \mu_i (x_j - x_r). \quad (24)$$

Interpretation. A high-influence agent can move price by directing attention to a fanatic. The effect is stronger when the fanatic’s view is more extreme.

3.11 Expected returns: value and momentum

Before revelation, expected return is:

$$\mathbb{E}_t[r(t+1)] = (1 - \pi)\Delta p_n(t+1) + \pi b(t), \quad (25)$$

where:

$$b(t) = x_r + u(t) - p(t). \quad (26)$$

Interpretation. Returns are predicted by a momentum component and a value component. Momentum comes from continued network diffusion. Value comes from possible revelation and reversion to fundamentals.

3.12 Short-term trading: riding and reversing the bubble

The short-term demand can be rewritten as:

$$d_i(t) = \frac{(1 - \pi)w_i}{\sigma_u^2}\Delta p_n(t+1) + \frac{\pi w_i}{\sigma_u^2}b(t). \quad (27)$$

Interpretation. The first term encourages rational traders to ride the bubble. The second term encourages them to trade against the bubble when valuation is too stretched.

4 Identification and instruments (what makes the model credible)

4.1 No empirical instrument

- The paper is a theory paper. It does not estimate a causal effect using an instrument, difference-in-differences design, or regression discontinuity.
- The GameStop episode is an illustration of the theory, not a formal identification exercise.
- The model is credible because it imposes strong structure: a fixed network, hardheaded and naive agents, mean-variance demands, and market clearing.

4.2 What disciplines the model

- **Belief discipline.** Beliefs follow transparent updating rules.
- **Network discipline.** Influence depends on the adjacency matrix and the echo matrix.

- **Market discipline.** Prices must clear the market.
- **Closed-form discipline.** Main objects have closed-form solutions, making comparative statics transparent.

4.3 Why the mechanism is not arbitrary

The paper does not simply assume that social media creates bubbles. It derives a precise chain:

- fanatics hold extreme views,
- naive investors learn gradually through the network,
- influencers change thought leadership,
- beliefs affect demand,
- demand affects price,
- rational traders may ride and then reverse the price movement.

4.4 Main alternative explanations

The paper abstracts from several forces that can also matter in real markets:

- strategic communication and pump-and-dump behavior,
- options-market feedback and gamma squeezes,
- short-sale constraints and short squeezes,
- endogenous network formation,
- changing attention based on past profits,
- brokerage frictions and platform outages,
- firm communication and central-bank communication.

Interpretation. These omissions are not failures; they define the paper’s clean experiment. The paper isolates the social-network channel.

4.5 Relation to existing theories

- Relative to DeGroot learning, the paper adds rational agents, fanatics, and asset-market equilibrium.
- Relative to persuasion-bias models, the paper shows how repeated social updating affects prices and portfolios.
- Relative to disagreement models of bubbles, the paper gives a network-based source of disagreement.
- Relative to behavioral theories of momentum and reversal, the paper links return predictability to social diffusion.

5 Tables and figures

5.1 Table 1: Model implications

L.H. Pedersen

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Table 1

Model implications. The first column contains the key elements of my model, showing how an investment idea starts in the model (A), initially affects prices (B), evolves over time (C,D), and how rationality ultimately sets in (E,F). The second column shows how these model elements capture the stylized evolution of historical bubbles, that is, the bubble anatomy of [Kindleberger \(2000\)](#) and [Shleifer \(2000\)](#). The third column shows how each element played a role in the GameStop case. Finally, the last column shows how these model elements help explain certain general asset-pricing effects in financial markets.

Model elements	Bubble anatomy	GameStop 2021	Asset pricing
A. Investors receive news, fanatics focus on one element	Initial displacement	Retail investors focus on plan to pivot online	Announcement effects, e.g., post-earnings drift
B. Short-term investors bet on network spillovers	Speculation	Some institutional investors are long	Momentum and fundamental momentum
C. Opinions spread through the network	Mania and emulation	More and more people hear about GameStop	Local bias and network spillover effects
D. Influencers follow a fanatic	Authoritative blessing	Elon Musk tweets a link to WallStreetBets	Excess volatility
E. Rational traders bet on reversal	Insiders sell out	Institutional investors sell	Value investing
F. Fundamentals are revealed or fanatics gradually learn	Crash: revelation or revulsion	Drops in January, March earnings announcement, etc.	Long-run reversal and value effect

Model elements, bubble anatomy, GameStop, and asset-pricing implications.

Read: The table maps model elements into bubble anatomy, the GameStop case, and broader asset-pricing effects. The sequence is: investors receive news, short-term investors bet on spillovers, opinions spread, influencers amplify fanatics, rational traders bet on reversal, and fundamentals are eventually revealed.

Economic message: The same framework explains both extreme events and regular asset-pricing patterns. GameStop is a highly visible version of a more general social-market mechanism.

5.2 Figure 1: Investor beliefs over time

Read: Naive investor beliefs rise gradually as optimistic views spread. With fully fanatic agents, beliefs converge to heterogeneous long-run limits. With nearly fanatic agents, beliefs eventually drift back toward rationality.

Economic message: Social learning can generate persistent disagreement. If fanatics never update, disagreement and mispricing can remain until revelation. If fanatics gradually learn, the bubble can peter out.

5.3 Figure 2: Price bubble and volume

Read: The price rises as the network transmits optimistic beliefs. It either crashes when fundamentals are revealed or slowly declines when fanatics become less extreme. Volume spikes early and then falls quickly.

Economic message: A bubble can remain even after trading volume falls. Volume is high when beliefs are changing; once beliefs stabilize, the bubble can survive with lower turnover.

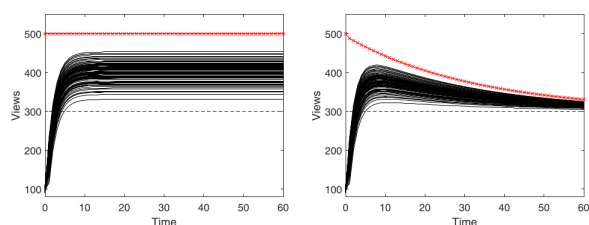


Fig. 1. Investor beliefs over time. The left panel shows how investors' beliefs about the fundamental asset value change over time in Example 2. The line on top (identified with cross-marks) shows the fanatic view, the dotted line shows the rational view, and the solid curves show the naive investors' views. Naive investors converge to different limits because of their differing social ties to fanatic or rational agents. The right panel shows the same, except that "fanatics" are not completely fanatic, they put 3% weight on the rational view.

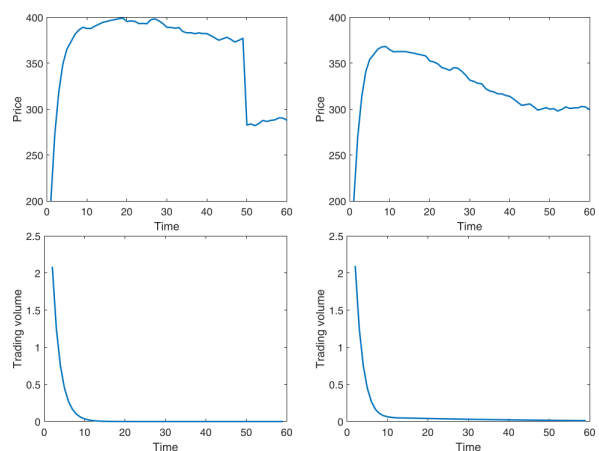


Fig. 2. Price bubble and volume. The top left panel shows the evolution of the asset price over time in Example 2 when fanatics are fully fanatic and the bubble bursts at time 50 when the fundamental value is revealed. The top right figure shows the price when fanatics put a small weight on the rational view, so the bubble peters out even though the fundamental value is not revealed over the shown time period. The bottom panels show the trading volume with fully fanatic agents (left) and fanatics who put a small weight on the rational view (right).

5.4 Figure 3: Short-term investor positions

Read: Rational short-term investors initially buy aggressively, later reduce exposure, and eventually move toward short positions as overvaluation becomes too large.

Economic message: Rational arbitrageurs do not necessarily short a bubble immediately. If social-network momentum is predictable, they may ride the bubble first and trade against it later.

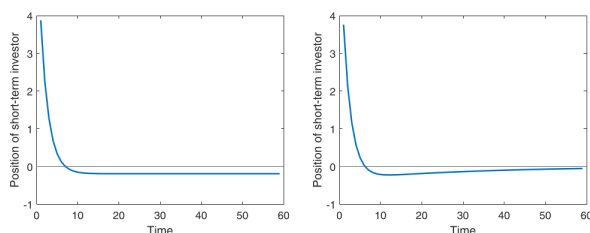


Fig. 3. Asset position of short-term investors. This figure shows the asset position of short-term investors over time in Example 2 assuming no revelation. Fanatics are fully fanatic in the left panel and nearly fanatic in the right panel. Short-term investors are initially long the asset, acting as momentum traders who ride the bubble. Later in time, short-term investors act as value investors who bet of a price reversion toward fundamentals when the bubble bursts.

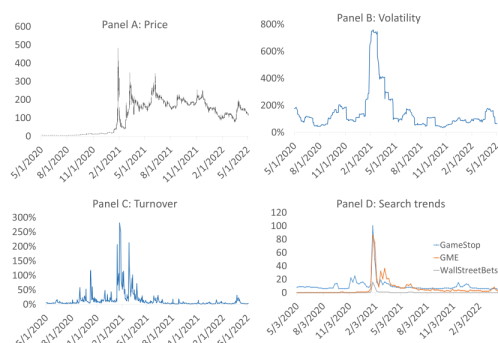


Fig. 4. GameStop price, volatility, turnover, and interest, 5/1/2020-5/3/2022. Panel A shows the GameStop stock price over time (open-high-low-close). Panel B shows the realized volatility (20-day annualized volatility). Panel C shows the daily turnover, and Panel D shows the social media interest as proxied by Google search trends for GameStop, GME, and WallStreetBets.

5.5 Figure 4: GameStop price, volatility, turnover, and attention

Read: GameStop's price rose dramatically in early 2021. The price spike coincided with extremely high realized volatility, very high turnover, and a large increase in search interest for GameStop, GME, and WallStreetBets.

Economic message: The case is consistent with the model's core prediction: contagious narratives can move attention, trading activity, volatility, and prices together.

6 Short summary

- Pedersen builds a tractable model of social networks and asset markets.
- Naive investors repeatedly average the views of people they follow.
- Fanatics are stubborn about extreme views.
- Rational investors become stubborn about truth after processing all initial information.
- Influencer value measures the effect of a naive investor's attention on long-run thought leadership.
- Thought leadership measures how much a hardheaded view matters for average beliefs and prices.
- The price equals a rational benchmark plus a network component.
- Long-run network mispricing equals fanatic mispricing weighted by fanatic thought leadership.
- Rational short-term traders may first ride network momentum and later bet on reversal.
- The model explains bubbles, volume spikes, excess volatility, momentum, reversal, social spillovers, and meme trading.

7 Conclusion

7.1 Core conclusion

Social networks can affect prices because investors do not only learn from public news and prices. They also learn from each other. When some agents are naive and others are stubborn, communication can create persistent disagreement and price distortions.

7.2 Economic intuition

- A stubborn view can persist.
- A naive network can repeatedly transmit that view.
- An influencer can amplify it.
- Trading demand converts propagated beliefs into prices.
- Rational short-term traders may initially reinforce the movement before eventually betting against it.

7.3 Takeaway

The paper changes the meaning of market power. In a social market, power is not only the size of a trader's wallet. It is also the ability to shape what others hear, repeat, and believe.